CAPITAL ALLOCATION / DECISION MEMORANDUM

Investment priorities for sustainable growth

To: Board of Directors, Tiny-Rod Hit Inc.

From: Senior Finance Manager

Date: January 2025 | Planning for May 2025 funding

RECOMMENDATION: PRIORITIZE PROJECT B

Approve a conditional $40 million envelope for the disruptive-technology program, released against technical, customer and intellectual-property milestones. Do not commit the full amount irrevocably before these gates are met.

Executive judgment

Project B best combines potential long-term economic value with the company's existing R&D capabilities and core-market knowledge. It can defend market leadership and create a differentiated platform at a lower initial capital requirement than Project A ($40 million versus $50 million). The 30% estimated failure risk is material; this is a disciplined innovation investment, not a low-risk extension of the current business. [2]

Under explicitly assumed cash-flow paths, Project A produces a modest positive net present value (NPV) of roughly $5–10 million and an internal rate of return (IRR) near 12%. Project B's probability-weighted NPV is roughly $25–30 million, with an IRR on the expected cash-flow stream near 18%, versus the 9% corporate WACC. These are directional planning estimates, not valuations inferred from revenue growth alone. The assumptions, sensitivities and cash-flow bridge appear on pages 2–3 and 7.

Why this choice is conditional

B's advantage depends on achieving scalable cash margins, timely commercialization and enforceable differentiation. A two-year launch delay can erase its modeled NPV within the fixed ten-year evaluation window. A offers valuable geographic and customer diversification, but limited local experience and new infrastructure needs create execution risks without much modeled return headroom.

If the Board requires both ventures

Earmark the full $100 million as $55 million for A and $45 million for B: $90 million of base project funding plus $5 million of contingency for each. Retain the $10 million contingency in cash until specifically authorized; do not force immediate spending simply because funds are available.

Timing convention: the assignment is a January 2025 decision but cites $100 million available as of May 2025. This report treats that amount as a prospective May planning assumption, not verified January cash. No deployment is assumed before Treasury confirms availability. [1]

SCOPE AND ASSUMPTIONS

1 | Decision basis and valuation discipline

Tiny-Rod Hit Inc. is described as consistently profitable, with a strong balance sheet and a healthy debt-to-equity ratio. These conditions support patient investment and funding flexibility, but do not substitute for positive project economics. The supplied corporate weighted average cost of capital (WACC) is 9%. [1]

Supplied project facts [2]	Project A: emerging markets	Project B: disruptive R&D
Initial funding	$50m upfront	$40m initial investment
Revenue outlook	20% annual growth for the first five years; 5% thereafter	If successful, over 50% annual growth for the first three years, then rapid penetration
Strategic fit	New customer segments; geographic and product diversification	Core-market leadership; competitive advantage; potential industry redefinition
Execution context	New product line, infrastructure and local partners; limited market experience	Strong internal R&D team and existing market expertise
Principal exposures	Geopolitics, regulation, local competition and currency	30% failure estimate; obsolescence, competition and IP protection

What the data do not establish

Revenue starting levels, margins, development timing, taxes, working capital, follow-on capex and terminal values are not supplied. Neither NPV nor IRR can be reliably derived from growth percentages alone. The financial case below therefore uses transparent, illustrative assumptions rather than claiming an investment-grade forecast.

Common valuation rules

Measure nominal USD, after-tax, unlevered free cash flow over ten years from funding. Year 0 is the May 2025 funding point; Year 1 is the next twelve months, not calendar 2025. Charge the full initial investment at Year 0 and place subsequent cash flows at year-end. Discount at 9%, with 12% and 15% hurdle sensitivities. No debt proceeds, interest payments or separate financing tax shields are added.

Exclude terminal value, salvage and working-capital recovery for both projects. This avoids allowing uncertain perpetual growth to drive the decision, but may understate durable platforms and can change the ranking. The supplied 5% long-run growth for A is applied only within the modeled period.

Illustrative operating assumptions—not supplied facts

A: $25m Year 1 sales, 20% growth through Year 5, then 5%; 20% free-cash-flow conversion. B: two development years with no incremental net cash flow; $25m launch sales in Year 3; 55% growth in Years 4–6, 20% in Year 7 and 10% in Years 8–10; 25% conversion if successful. B’s growth timing interprets the reference as the first three post-launch growth periods.

Conversion means cash remaining after tax, recurring capex, working capital and ongoing reinvestment. Initial budgets must cover all prelaunch build/R&D and funding needs; any omitted losses or later investment reduce value. Earlier staging is a proposed safeguard, not a benefit included in the base valuation.

FINANCIAL ASSESSMENT

2 | Directional returns and implications

The ranges below communicate approximate scale, not statistical confidence intervals. They depend on the illustrative assumptions on page 2 and the rounded cash-flow bridge on page 7.

Case at 9% WACC	Indicative NPV	Indicative IRR	Interpretation
A: illustrative operating case	+$5–10m	~12%	Positive, but limited cushion over the 9% hurdle.
B: successful outcome	+$55–60m	~24%	Strong potential; conditional on technical and commercial delivery.
B: failure outcome	−$40m	Not meaningful	No significant revenue or salvage; full initial investment lost.
B: 70% success / 30% failure	+$25–30m	~18% on expected cash flows	Probability-weighted NPV; not the return an individual outcome will necessarily earn.

How to read the comparison

B's expected NPV weights the successful inflows at 70%, assumes zero failure inflows and charges the full $40m cost in all cases. Its expected-stream IRR is not an average of successful and failed IRRs. A has no supplied failure probability, so its case is not equivalently probability-adjusted; country and execution losses remain to be underwritten.

Revenue growth is not IRR. Even A's 20% early sales growth produces only a roughly 12% project IRR after capital needs and cash conversion. B's larger modeled NPV is more relevant to shareholder value than its higher percentage return alone.

Downside tests: assumptions matter more than precision

One-at-a-time test	A: approximate NPV	B: probability-weighted NPV
All operating cash inflows 25% lower	−$5–10m	+$10–15m
Discount rate raised to 12%	Around zero	+$15–20m
Discount rate raised to 15%	−$5–10m	+$5–10m
B launch delayed two years; same Year 10 cutoff	Not tested	Around zero

Tests are separate, not a combined forecast. Delay shifts B inflows two years later and excludes those beyond Year 10. Higher discount rates test residual country, market and execution risk; they are not an automatic second charge for the already-modeled binary failure event.

Decision thresholds

A needs roughly 17% cash conversion, versus the assumed 20%, just to break even at 9%. B needs about a 40% success probability, versus the supplied 70% implied success case, if its successful cash-flow path is achieved. These thresholds are highly conditional: reducing B's successful inflows or delaying launch raises its required success probability. Validate the cash economics before treating the apparent headroom as real.

RISK OWNERSHIP AND CONTINGENCY

3 | Project B: three priority risks

The Board should fund evidence, not simply elapsed time. The following proposed triggers are management controls, not facts from the reference. They are intended to limit irreversible spending while protecting the ability to scale a successful platform.

1. Technical failure and obsolescence | Owner: CTO

Exposure: The supplied 30% failure estimate means the $40m could generate no significant revenue. Even a working prototype can be overtaken by a competitor before launch.

Mitigate: Use an initial $10m feasibility tranche, followed by two $15m releases. Define measurable performance, reliability and integration benchmarks against the current customer alternative before work begins. Require independent technical review, quarterly competitor scans and modular architecture that allows replacement of vulnerable components.

Trigger and contingency: If a critical benchmark is missed at two successive gates, or the technical edge no longer supports customer willingness to pay, freeze new commitments. Allow a time-boxed redesign only with an approved positive incremental NPV. Otherwise terminate within 90 days, cancel cancellable contracts and redeploy the team. Reuse or license IP where practical, but assume no salvage in the approval case.

2. Weak adoption, cash conversion or cost escalation | Owners: CFO / CCO

Exposure: High sales growth may not translate into cash if pricing falls, incumbent responses intensify, the new product cannibalizes existing profit or follow-on spending exceeds the $40m scope.

Mitigate: Obtain at least two paid or contracted lighthouse customers before scale-up; test pricing and support costs in pilots. Use existing sales and service channels. Build an incremental cash model net of cannibalization and all future development, launch and working-capital needs. Monitor forecast-at-completion and cash conversion monthly.

Trigger and contingency: Re-underwrite if signed demand stays below 70% of the approved plan for two quarters, expected mature cash conversion falls below 20%, or the base cost forecast exceeds $40m. Reduce launch scope and fixed commitments, consider licensing or bundling into core products, and release contingency only if incremental value remains positive. Do not borrow to preserve an obsolete plan.

3. IP infringement or loss of protection | Owner: General Counsel

Exposure: An injunction, unclear ownership or copying can delay launch, raise licensing costs and undermine the technology’s pricing power.

Mitigate: Complete independent freedom-to-operate review before the prototype gate and again before commercialization. Secure employee and partner IP assignments, selective patent filings, trade-secret access controls and contractual rights to critical software and components.

Trigger and contingency: A material unresolved claim or critical ownership gap blocks the next tranche and the affected release. Evaluate a design-around or a license capped by its incremental economic value. Isolate the disputed feature; abandon it or the project if a clean commercial route cannot be established. Escalate material legal exposure immediately to the Board.

Staging improves loss control only if commitments are genuinely deferrable. If the full $40m must be irrevocably committed upfront, return to the Board before proceeding; the base valuation already assumes that full exposure.

CONDITIONAL PORTFOLIO MANDATE

4 | If both proceed: allocate the $100m

Assuming both projects pass diligence and are viable, fully earmark the $100m across the two ventures without spending it all at inception. The allocation preserves each project's stated base requirement rather than assuming either can deliver its full strategy on a reduced budget.

Allocation ($m)	Project A	Project B	Total
Base project funding	50	40	90
Project-specific contingency held in cash	5	5	10
Total Board envelope	55	45	100

The larger allocation to A reflects its larger upfront capital need, not a reversal of the preference for B. The $5m reserves are proposed spending caps, not estimated extra costs. They remain in Treasury-controlled cash, require CFO and Board authorization and may be reallocated only by explicit Board approval.

Sequence spending to preserve choices

Stage	Funding and evidence required
Before May deployment	Treasury confirms the $100m and operating liquidity; legal and operational diligence clears the chosen markets and B's initial scope. No premature commitments.
Funding / launch preparation	Provide A's full $50m at its launch decision after partner, regulatory, FX and infrastructure plans are approved. Release only B's first $10m, if commitments can be staged.
Subsequent B gates	Release $15m after independently validated technical proof and IP clearance; release the final $15m after pilot/customer validation and a refreshed positive investment case.
Contingency release	Keep $5m for each venture unspent. Fund only identified incremental value or orderly risk containment; escalate any need beyond the $55m / $45m envelopes.

Portfolio value beyond stand-alone returns

A broadens geography and customer exposure, while B renews the core franchise and creates technology options. Existing capabilities can support B; local partnerships can build A's route to market. Maintain separate accountable teams and a shared portfolio steering committee to avoid competition for scarce engineering and commercial staff.

Diversification is useful but not guaranteed: both remain exposed to technology demand, competition and execution capacity. Do not assume independent risks, cross-project subsidies or unproven synergies. On the illustrative cases, combined NPV is roughly $35–40m before extra spending; consuming all $10m of contingency early would reduce it to roughly $25–30m, with no assumed offsetting benefits.

Balance-sheet guardrail

Use cash for the base plan; do not add debt simply to improve headline equity returns. Confirm that the $100m is surplus to operating cash, existing obligations and any covenants. If it is not, delay A rather than silently underfund its $50m launch. A committed facility can bridge timing mismatches, subject to liquidity and leverage limits; no new debt capacity is assumed from the qualitative "healthy" ratio alone.

APPROVAL AND ACCOUNTABILITY

5 | Board actions and management controls

Resolutions requested

1. Primary choice: approve B's conditional $40m envelope and the three risk controls on page 4. Authorize only the initial $10m once cash, scope and milestones are confirmed. Retain the remaining $60m of the prospective cash balance uncommitted under this single-project recommendation.

2. Dual-project alternative: if the Board requires both, approve the $55m / $45m envelopes on page 5, including the $10m held contingency. Require separate investment cases and do not treat a good aggregate return as permission to fund a negative-value component.

3. Delegation: appoint the CFO as portfolio sponsor, the CTO as B's delivery owner and an international operations executive as A's owner if approved. Reserve material scope changes, reserve draws, new borrowing and any envelope overrun for Board decision.

Evidence required before major commitments

Decision checkpoint	Required evidence / decision
Before first release	CFO reconciles the May cash outlook to operating liquidity, covenants and other commitments; CTO confirms what the $40m does and does not fund. Obtain a costed resource plan.
B technical / IP gate	Independent benchmark results, freedom-to-operate opinion and an updated competitor assessment. Proceed only if differentiation remains commercially relevant.
B commercialization gate	At least two paid or contracted lighthouse customers, validated pricing and support economics, incremental cash flows net of cannibalization, and refreshed NPV / downside analysis.
A launch gate, if applicable	Country-level regulatory and tax review; partner diligence; infrastructure and working-capital budget; local competition, currency and repatriation plans. Stage country entries only if the $50m scope can genuinely support it.

Reporting, stop rules and reallocation

Management should submit a monthly dashboard of cash spent and committed, forecast cost-to-complete, technical milestones, contracted demand, cash conversion, IP issues and staffing capacity. Refresh the portfolio NPV and liquidity stress test quarterly; distinguish sunk costs from future incremental decisions.

Stop or redesign B when a gate fails and no positive incremental case remains. Release uncommitted cash back to Treasury rather than automatically transferring it to A. Approve A only on its own economics: the base case is only slightly above the hurdle and near break-even at a 12% discount rate.

Why B remains the first choice—and when A could win

B offers the stronger modeled value contribution, lower initial capital exposure and better fit with existing expertise. A's diversification is strategically valuable, but local learning, regulation and FX risk weigh on its thin return cushion. A could become preferable if validated local economics or partnerships improve its cash generation, or if B's customer, technical or IP evidence reduces its expected value below A's. Neither a strong balance sheet nor a strategic label justifies ignoring negative incremental NPV.

DIRECTIONAL ESTIMATES, NOT A DETAILED MODEL

Appendix | Valuation bridge and sources

Illustrative annual cash-flow bridge ($m, rounded to the nearest million). Revenue is shown only to make the assumed cash conversion transparent. Calculations use the underlying growth paths before rounding; the report intentionally presents NPV and IRR only at directional precision.

Year	A sales	A free cash flow	B sales: success	B free cash flow: success
0	—	−50	—	−40
1	25	5	0	0
2	30	6	0	0
3	36	7	25	6
4	43	9	39	10
5	52	10	60	15
6	54	11	93	23
7	57	11	112	28
8	60	12	123	31
9	63	13	135	34
10	66	13	149	37

Calculation logic and limitations

NPV = discounted annual free cash inflows less the full Year 0 investment. IRR is the discount rate that makes that cash-flow stream's NPV zero. For B, use 70% of each successful operating inflow and zero failure inflows, while retaining the full $40m initial outflow. That produces the expected-stream IRR; no meaningful conventional IRR exists for the all-loss failure outcome.

The unweighted successful inflow PV for B is roughly $95–100m at 9%, versus about $60m for A. B's break-even success probability is therefore roughly $40m divided by $95–100m, or about 40%. For A, scale its assumed 20% conversion down until the inflow PV equals $50m to obtain the roughly 17% break-even conversion.

This is not a full probability distribution. The reference's 30% failure estimate is interpreted as a binary outcome; partial successes, correlated macro shocks and additional funding rounds could materially lower value. No reserve spending, staging benefit, synergy, terminal value or salvage is included in the base cases. A longer horizon or terminal value could change the ranking.

Sources and attribution

[1] Board assignment supplied with this request: company financial context, $100m cash as of May 2025, 9% WACC, January 2025 planning date and Board directives.

[2] Tiny_Rod_Hit_Inc_Reference.pdf, p. 1, supplied reference: project investment requirements, growth descriptions, 30% B failure estimate, strategic fit, capabilities and principal risks.

All other numerical inputs, including sales levels, cash conversion, launch timing, ten-year horizon, discount sensitivities, allocation reserves and management triggers, are proposed planning assumptions developed for this report. No external market forecasts or valuation comparables are used.